standards. The drone manufacturers will be looking at the possibility of manufacturing efficiency in the supply chain and scalability in their target market before deciding to invest in their Bill of Material (BOM) and developing their network of suppliers in order to meet the required compliances.

While the policy has plans to allow 100% FDI (Foreign Direct Investment) in the manufacturing of Remotely Piloted Aircraft (RPA), the foreign manufacturers should be willing to change their design and meet the NPNT compliances

UAV Types	The Indian Drone Policy classifies drones based on their total weight with cargo and fuel
Nano drones	250gm
Micro-drones	250gm to 2kg
Small drones	2kg to 25kg
Medium drones	25kg to 150kg
Large drones	Upwards of 150kg
Manufacturing Standards	Airworthiness - specific hardware standards India is defining to certify a drone as reliable
	Bimodal control - allowing air traffic to take control of a drone in BVLOS flight
Mandatory Features for Drones in India	GPS
	Return-to-home (RTH)
	Anti-collision light
	ID plate
	A flight controller with flight data logging capability
	RFID and SIM/No Permission No Takeoff (NPNT)
Drones for Security and Surveillance	Camera resolution
	Artificial intelligence
	Multiple GPS for redundancy
	Integrated application software with a geographic map
	Real-time video transmission
Standard Packaging	Drones
	Payloads
	GCS
	Kits
	Accessories

Component and sub-system level development in India to play the pivotal role in innovation and subsequent commercialisation

Currently, more than 80% components are imported, which poses the biggest challenge in front of the domestic manufacturers of drones. The supplier base of the battery packs, which being one of the most critical components in the system, is having poor presence in India and it requires to be enhanced.

Component level innovation encompassing the miniaturisation of sensor technology, low-cost 3D printing technology and assembly of kits holds the key and has the potential to provide an impetus to the continuous evolution of the design and development of low-cost, high-value UAVs. Sensors including hyperspectral imager, SAR (Synthetic Aperture Radar) and LiDAR (Light Detection and Ranging) are going to play the role of a most critical hardware component, which can help drone industry revolutionise agriculture, search and rescue, town and country planning, etc.

Sub-system development/component level R&D is going to play a strong role in increased commercialisation of innovative technologies in the drone industry. For example, a strong understanding BLDC specs, sensors, and propellers can be considered to be the first level of development. Value addition to the same can lead to innovative products on sub-system and system level, opening the doors for patents and further commercialisation.

Additionally, Indian landmass offering the most varied terrain across the nation will demand for high altitude engines in order to meet the varying climatic conditions.

The proponents of the technology are driving the technology innovation through the integration

Developing a Sustainable Drone Technology	Embracing the Big Data
• Technological advancement in the field of drone applications is going to take a big leap through the integration of artificial intelligence, machine learning, virtual reality, augmented reality, Internet of Things (IoT), GPS, 5G, etc. • The consumers and enterprises across public and private sectors will be looking for a one-stop solution that leverage a spectrum of technologies to create an automated, secure and economical solutions.	• A classic example of technology integration can be seen in the development of predictive analytics solutions for various commercial purposes by using drone sensors, artificial intelligence, and machine learning technologies. • Manufacturers of drones are working to develop drone systems that can land and take-off automatically, while avoiding collisions, with the help of drone sensors. • Such drones can fly reliably and repeatedly without the pilot.